

Skytree Installation

Release 15.4.0

SKYTREE®

Notices

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Customers with a support contact can contact support using any of the following methods:

- Log in to support.skytree.net.
- Send questions to support@skytree.net.
- Call 408.392.9300 and select option 3.
- You can also view additional Skytree documentation by logging on to support.skytree.net.

Freemium customers can send inquiries to the LinkedIn® Community Group for Skytree:

<https://www.linkedin.com/groups/4468542>

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Overview

This document describes how to install Skytree on non-edge-node HA clusters using the `skytree-installer.tar.gz` scripts and will guide you through the following steps:

1. Configure a installation properties file `skytree.config`. The installers will refer to the variables/properties in this file.
2. Install Skytree ML/Core software on all node manager nodes and on the edge node (configurable).
3. Install Skytree Platform software (along with tomcat and database) on the edge node.
4. Start the Skytree platform service and run a smoke test.

Requirements

This section describes the hardware and software requirements for Skytree. Refer to the following for more specific information:

- [Hardware Requirements](#) (page 1)
- [Software Requirements](#) (page 1)
- [Cluster Configuration](#) (page 2)
- [Supported Browsers](#) (page 3)

Hardware Requirements

Below are the minimum and recommended hardware requirements for each node in your cluster.

- **Minimum:** Your hardware must have at least 16 physical cores with 128 GB of RAM (DDR3 or faster).
- **Recommended:** We recommend at least 40 physical cores with at least 320 GB of RAM (DDR3 or faster).

Software Requirements

The following are assumed to be already available on your system:

Operating System

The operating system must be one of the following:

- RHEL 6.5
- CentOS 6.5

YARN Platform

Be sure to verify that YARN commands can be run on the node that will include Skytree Platform. Supported YARN platforms include:

- Hortonworks Data Platform 2.1.x, 2.2.x, or 2.3.x with Zookeeper
- Cloudera 5.4.x or 5.5.x with Zookeeper
- MapR 4.0.2 with Zookeeper
- Amazon EMR Hadoop 2.4/AMI 3.8

Curl

In order to use the Skytree REST API, you will need a terminal with curl to implement the HTTP requests. Refer to curl.haxx.se for more information.

Java

Java 1.7 or 1.8 is required to be installed on all nodes.

Cluster Configuration

Important Guidelines

Please keep the following in mind when installing Skytree on the Hadoop Cluster:

- Skytree Platform and accompanying software should be installed on an edge node. This edge node must not be running Ambari or Cloudera Manager.
- It is also HIGHLY recommended that Hadoop services such as datanode, namenode, nodemanager and resource manager are not running on this edge node.

Port Configuration

Refer to the tables below for a list of ports that are used by Skytree as well as those for the Hortonworks and Cloudera Hadoop distributions.

Note: Contact your Skytree, Inc. Sales Engineer if the following required ports are already used and/or if you require a different set of available ports,

Table 1: Skytree Port Configuration

Port Num/ Range	Property	Description	Accessed by External Host?
9001 - 9010	skytree_ports	Comma-separated list of ports used by Skytree Server to communicate with Skytree Platform	Yes
9090	skytree_ipc_callback_server_port	Used by Skytree Server to send machine learning operation status updates to Skytree Platform	Yes

Table 1: Skytree Port Configuration (continued)

Port Num/ Range	Property	Description	Accessed by External Host?
9095	spark_launcher_server_port	Port used by Skytree Platform to communicate with the Skytree Spark ETL Launcher daemon	No
9096	spark_status_server_port	Port used by the Skytree Spark ETL Launcher daemon to communicate status updates back to Skytree Platform	No
9097	skytree_launcher_port	Port used by Skytree Platform to communicate with the Skytree Server Launcher daemon	No

Table 2: Hortonworks Default Ports

Skytree Property	Sample Value
fs_default_name	hdfs://<namenode_host_name>:8020
yarn_resourcemanager_address	<yarn_resourcemanager_host_name>:8050
skytree.zookeeper.connection	<zookeeper_quorum_peer_hostname>:2181

Table 3: Cloudera Port Configuration

Skytree Property	Sample Value
fs_default_name	hdfs://<namenode_host_name>:8020
yarn_resourcemanager_address	<yarn_resourcemanager_host_name>:8032
skytree.zookeeper.connection	<zookeeper_quorum_peer_hostname>:2181

Tomcat Properties

By default the Tomcat server will be set up on port 8080 (or 8443 for ssl). Refer to the Tomcat documentation for instructions on how to change this port, if desired.

Supported Browsers

The following table lists the browser/OS combinations that are supported in Skytree. The supported browser version is assumed to be the current browser version.

Table 4: Supported Browsers

Browser	OS
Google Chrome™	Windows® 7, Windows 8, Mac OS X®

Table 4: Supported Browsers (continued)

Browser	OS
Mozilla Firefox®	Windows 7, Mac OS X

Before You Begin

The Skytree installation script will prompt you to provide certain information during the installation process and to make changes to the `skytree.config` properties file. We recommend that you keep this information nearby during the installation process.

Information Required During Installation

Below is a list of the items that you will need to know before completing the installation.

- The Hadoop distribution (for example, hw228, hw23, cd54, etc.)

- The user name and password for the Skytree download site

- The host name of one of the servers running Zookeeper (FQDN)

- The Zookeeper port number (typically 2181)

- The path to libhdfs.so (for example /usr/hdp/2.3.2.0-2950/usr/lib/libhdfs.so)

- The Hadoop configuration directory where the `yarn-site.xml` is located (for example, /etc/hadoop/conf)

- The JAVA HOME path, i.e., the absolute path where Java is installed on all the node manager nodes as well as the data nodes. These must be the same and be a supported version of Java.

Optional: If Skytree jobs will run on a specific YARN queue then provide the name of that YARN queue.

Additional Input Information

The following additional input information will be required during the installation:

1. We assume that passwordless or key-based ssh is possible to these nodes from the edge node. If ssh is key based, then the key file must be available on the edge node.
2. The Skytree license file (`skytree.lic`) must be copied onto the edge node. This file was provided via email. Contact support@skytree.net you need assistance retrieving this file.

Installing Skytree

Perform the following steps to install Skytree. Note that as a result of this installation, Postgres 9.4 will be installed on the edge node.

1. Download the `skytree-installer.tar.gz` package on the edge node.
2. Open a terminal window or command prompt and navigate to the location where you saved the downloaded file.
3. Unzip the `skytree-installer.tar.gz` file using the following command:

```
sudo tar -xzf skytree-installer.tar.gz
```

4. CD into the "installers" directory just created.
5. Download the Skytree binaries using the following command on the edge node.

```
sudo ./download_skytree_binaries.sh
```

You will be prompted to provide the following information:

- The Hadoop distribution (for example, hw228 or cd54)
- The user name for the Skytree download site
- The password for the Skytree download site

This script downloads the following 12 files into a folder called `skytree-binaries`:

- `SkytreeDB-<version>.x86_64.rpm`
- `SkytreeServer-<version>.-rel_99116.el6.x86_64.rpm`
- `skytree-platform-<version>.506d127_99229-1.x86_64.rpm`
- `SkytreeDepLibsCompleteZipSrc-1-0.x86_64.rpm`
- `skytree-python-sdk-<version>.tar.gz`
- `spark-1.5.1.tgz`
- `postgresql94-9.4.1-1PGDG.rhel6.x86_64.rpm`
- `postgresql94-contrib-9.4.1-1PGDG.rhel6.x86_64.rpm`
- `postgresql94-libs-9.4.1-1PGDG.rhel6.x86_64.rpm`
- `postgresql94-server-9.4.1-1PGDG.rhel6.x86_64.rpm`
- `context.xml`
- `catalina.properties`

6. Uncomment and configure the following properties in the `skytree.config` file.

```
#skytreelibhdfs=  
#zookeeper=  
#zookeeperport=2181  
#hadoop_dir_conf=  
#spark_home  
#skytree_java_home=
```

```
#distro=
```

- Optional: If a specific YARN queue is to be used uncomment and update the following property

```
#skytree_yarn_queue=
```

- Run the `copy_install_data_nodes.sh` script on the edge node. This will install the Skytree ML/Core software on all node manager nodes.

```
sudo ./copy_install_data_nodes.sh --help
```

You will be prompted to provide the following information:

- The location of the Skytree license file
- The folder where the binaries were downloaded
- The username (with sudo privileges) to be used to SSH/SCP to the nodes/hosts
- OPTIONAL: A key file to be used to SSH/SCP

If a prior version of Skytree core already exists on the node manager nodes, then an upgrade option will be available.

Note: If a key file is not provided, then passwordless SSH is assumed from the current node to all other nodes on which the installation is to be performed.

- Run the `install-skytree-platform.sh` script on the edge node using the following command:

```
sudo ./install-skytree-platform.sh
```

You will be prompted to provide the following information:

- The folder where the binaries were downloaded
- The location of the Skytree license file

When the installation is complete, the next steps are to start Skytree and run a validation test. To do this run: `sudo ./starttestskytree.sh`. This will start the "skytree-platform" service and run smoke tests. Bear In mind this may take 5-10 minutes to complete as it is loading some test data and running some small test models. However, additional properties may need to be set first if your environment uses Kerberos, Impersonation, or LDAP. Refer to the next sections.

Environment Information for Kerberos, Impersonation, and LDAP

This section describes some additional information about setting up your environment for Kerberos, impersonation, and LDAP before starting Skytree.

If you are not running Skytree in a Kerberos environment and/or if your environment does not use LDAP authentication, then you can skip this section.

For Kerberos Hadoop Environments

The following information is provided for users with Kerberos integrated into the environment:

- A keytab file must be generated for the skytree service account (from the skytree_user property). This keytab file must be present in the same location on all nodes on the Hadoop cluster including the Edge node. Make sure that the skytree user has read-write access to this file.
- A “skytree” user principal must be created in the same realm as the Hadoop cluster. The Apache Tomcat server must run as the “skytree” user principal. The “skytree” user must exist on all nodes of the cluster.
- You must set the skytree_user_keytab_location property in the skytree.properties file.
- Skytree Platform will maintain its own workspace under the /user/skytree folder in HDFS. The “skytree” user principal must have read-write access to this folder.

Impersonation

- Skytree Platform uses secure Hadoop impersonation to access any source files that are located outside of the “skytree workspace” and that are not accessible by the “skytree” user principal.
- Secure Hadoop impersonation requires a change to the core-site.xml configuration file: the “hadoop.proxyuser.skytree.groups” and “hadoop.proxyuser.skytree.hosts” properties must be added to core-site.xml and configured with the appropriate values. After this file is changed, you may need to restart YARN and the HDFS cluster.
- User accounts created in Skytree Platform will need to be associated with either a Kerberos principal or a Linux login for impersonation to work correctly. This association can be set up during user account creation.

LDAP

If your environment uses LDAP authentication, then the following additional properties must be set in the /etc/skytree/skytree.properties file.

```
# LDAP configuration properties (values required only if LDAP is configured)
skytree_authentication_type=ldap
skytree_ldap_host=ldap.example.com
skytree_ldap_port=389
skytree_ldap_base_dn=ou=employees,dc=example,dc=com
skytree_ldap_email_attribute=mail
skytree_ldap_username_attribute=cn
skytree_ldap_display_name_attribute=displayName
# The dn to do user lookups.
skytree_ldap_bind_dn=cn=skytree-plaform,ou=app-auths,dc=example,dc=com
# The password of the bind dn.
skytree_ldap_bind_pw=password1!
skytree_ldap_default_realm=example.com
# If using LDAP with SSL/TLS, set the following to true
# skytree_ldap_use_start_tls=true
```

Note: The skytree_ldap_email_attribute and skytree_ldap_default_realm properties allow for cases when the AD “mail” attribute does not contain a valid e-mail address for the user. In that case, uncomment these and change the skytree_ldap_email_attribute and skytree_ldap_default_realm properties to appropriate values. If the AD “mail” attribute contains a valid e-mail address, then these can remain commented out.

Note: If you are using a self-signed certificate, then the certificate must be imported into the JDK (being used by Tomcat) keystore.

Run the Validation Test

To start and smoke test Skytree on Tomcat, run the following command from the `/opt/skytree/platform/tests/` directory (or as root):

```
$ sudo cd /opt/skytree/platform/tests/  
$ sudo ./start_test_skytree.sh
```

This script takes an optional argument: the URL for tomcat. This defaults to `http://localhost:8080`.

If you have not changed the default port for Tomcat, you can run the script with no argument. If you have changed the default port for Tomcat, or you want to check access through DNS or IP locally, you can pass in an alternate URL. For example:

```
$ sudo cd /opt/skytree/platform/tests/  
$ sudo ./skytree_tests.sh "http://127.0.0.1:8888"
```

The smoke test normally completes after 5 minutes. Once complete, you should see output similar to the following.

```
Running Skytree REST Api smoke tests for host URL http://localhost:8080  
[TestNG] Running:  
Command line suite  
.  
....  
=====  
Command line suite  
Total tests run: 5, Failures: 0, Skips: 0  
=====  
Completed tests!
```

What's Next?

After Skytree is installed and the validation test has run successfully, you can begin preparing data and training and testing models using Skytree. Refer to the Skytree documentation for more information on configuring Skytree through the REST API, the Python SDK, or the GUI. The documentation is available by logging on to support.skytree.net.